When Post-Editing Helps and When It Harms: Evaluating LLM Post-Editing of NMT for Tagalog and Spanish

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California State University, Dominguez Hills
Research assistant: Shazan Ansar Mohammed; principal investigator: Benyamin Ahmadnia

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***Abstract*—Large language models (LLMs) can post-edit neural machine translation (NMT) drafts, but an editor that repairs errors can also change what a draft already rendered correctly. We evaluate this trade-off for English–Tagalog and English–Spanish in both directions on the held-out FLORES+ devtest set (1,012 sentences per direction). A sentence-split NLLB-200 3.3B system supplies the draft; Llama 3 8B Instruct and GPT-4o Mini edit it zero-shot, with one retrieved development exemplar, or with a single-call reason-then-edit format. Prompts, output contracts, and analyses were frozen on development data before the test set was opened, and all 24,288 outputs are evaluated. GPT-4o Mini improves COMET-22 in all twelve conditions (+0.26 to +1.00 points, all significant after Holm correction), most for the Tagalog pair and with the retrieved exemplar. Llama 3 8B never improves significantly; it is significantly worse in six conditions, and its reason-then-edit outputs degrade 247 of 1,179 strong drafts (20.9%), against at most 4.3% for any GPT-4o Mini strategy. Editing helps weak drafts almost everywhere, whereas strong drafts are at best preserved. Average quality, preservation of strong drafts, and output validity should therefore be assessed separately before an LLM post-editor is deployed.**

***Index Terms*—automatic post-editing, machine translation, Tagalog, Spanish, over-editing, LLM evaluation**

I. Introduction

An automatic post-editor must decide both what to correct and what to leave alone. A fluent rewrite can repair an omission, but it can also alter a name, number, or source relation that the machine translation (MT) draft already expressed correctly. When drafts are strong, mean quality changes therefore give an incomplete account of editing behaviour. LLMs can improve neural MT output, yet they also produce hallucinated edits [1].

We study this tension for English–Tagalog and English–Spanish in both directions. A fixed NLLB-200 3.3B system [2] produces sentence-split drafts, and two instruction-following editors, open-weight Llama 3 8B Instruct and commercial GPT-4o Mini, receive the source and the draft. The two language pairs provide a limited resource contrast; language, direction, domain, and baseline quality remain intertwined, so we report pair and direction differences descriptively rather than as causal resource effects. We ask: **RQ1:** How do editing gains and harms differ between the Tagalog and Spanish pairs, conditional on baseline quality? **RQ2:** How do the three prompting strategies compare within each model and direction? **RQ3:** When do editors change already strong drafts and degrade their measured quality or source fidelity? **RQ4:** How do effects differ between the two directions of a language pair?

This is an empirical evaluation, not a new post-editing architecture. Anchoring an LLM editor on a frozen NMT draft and selecting in-context exemplars by semantic similarity follow established prior work [3], [4], and we do not claim either as a contribution. Our contributions are:

- a held-out evaluation in a new empirical setting (four Tagalog/Spanish directions) against a stronger, sentence-split NLLB-200 baseline, with the protocol frozen on development data before the test partition was opened;
- a two-editor comparison (open-weight and commercial) across three prompting strategies, with paired confirmatory tests and Holm multiplicity control;
- explicit failure accounting: every planned output is retained, and output-contract failures are reported as outcomes rather than discarded;
- an over-editing and preservation analysis of strong drafts that separates metric-defined harm from heuristic preservation screens.

II. Related Work and Contribution Boundary

LLM post-editing. Raunak et al. [1] formalize direct translation post-editing with GPT-4 and report both quality gains and hallucinated edits. Ki and Carpuat [5] guide LLM post-editing with external error annotations, and xTower [6] trains a model to explain and correct translation errors, with expert assessment of its explanations. Our reason-then-edit condition differs: an unmodified editor writes its own brief diagnosis in the same call, and we treat that text as an observable output, not as a verified explanation. LangMark [7] provides human post-edit triplets; FLORES+ references are independent translations, so our reference-based gains are quality proxies and do not measure reduced human editing effort.

Exemplar selection. Vilar et al. [4] analyse prompting and example selection for translation. We retrieve one development exemplar with LaBSE [8]; the similarity is semantic, and one deterministic exemplar is a reproducibility choice, not a claim of optimality.

Relationship to our earlier ICAI study. Ahmadnia et al. [3] study English-to-German test-time domain adaptation with a frozen OPUS-MT draft, GPT-4 Turbo, multilingual Sentence-BERT top-three exemplars, and an optional second terminology-verification call. Frozen-NMT anchoring and semantic exemplar retrieval are shared with that work and are cited here as established methodology. The present study changes the empirical setting (four Tagalog/Spanish directions, FLORES+), the baseline (sentence-split NLLB-200 3.3B), and the editors (one open-weight and one commercial), uses a single-call reason-then-edit instead of a second verification call, and adds explicit failure and over-editing analysis. Because data, models, and metrics differ, we make no numerical comparison across the two papers.

Evaluation. We use COMET-22 [9], [10] as the primary metric, with BLEU [11] and chrF++ [12], [13] reported with full signatures [14]. No scalar metric replaces source inspection, so preservation screens are reported alongside quality.

III. Method

NMT baseline (S0). Drafts come from facebook/nllb-200-3.3B (FP16; four beams, no sampling, early stopping, length penalty 1.0, maximum length 512; inputs checked rather than truncated). The primary baseline segments each source with pysbd 0.3.4 (character spans, `clean=False`; English rules for Tagalog as a documented heuristic, with title, clock, *atbp.*, and parenthetical safeguards), translates multi-sentence items fragment by fragment, and joins the fragments with one space; single-sentence items reuse the unsplit hypothesis. Segmentation is lossless by construction. The unsplit system is kept as a diagnostic, outside the confirmatory tests.

Editors. We use the original Meta-Llama-3-8B-Instruct checkpoint (BF16, vLLM 0.11.2, pinned revision) [15] and the dated gpt-4o-mini-2024-07-18 snapshot [16]. Both decode with temperature 0, top-p 1, seed 42, and a 1,024-token output cap. Every request carries the same system instruction: the source is authoritative; source-supported names, numbers, dates, and meaning must be preserved; stylistic rewriting is discouraged; and an accurate, natural, grammatical draft must be returned unchanged. Source and draft are passed as delimited data.

Strategies. *Zero-shot* returns one JSON field, `edited_translation`. *Retrieved one-shot* adds one solved development example (its source, NLLB draft, and reference) before the query. *Reason-then-edit* returns a free-text `diagnosis` followed by `edited_translation` in a single call; only the translation is scored. The current query's reference never enters any prompt.

Retrieval. LaBSE [8] embeds all 997 development sources per source language (normalized, 768 dimensions); an exact FAISS inner-product index returns the most similar development item, excluding exact duplicates of the query source and breaking ties by the smaller sentence ID.

Output contract and fallback. Structured decoding (vLLM; OpenAI strict JSON schema) constrains the response shape, and a strict validator then requires valid JSON without duplicate keys, the exact field set and order, non-empty fields, and a normal stop. Otherwise, the raw response is retained and the system output is the unchanged baseline draft:

$$\hat{y}_i = \begin{cases} e_i, & \text{if the response passes the contract,} \\ b_i, & \text{otherwise.} \end{cases} \tag{1}$$

Failures are neither repaired nor selectively regenerated, so every evaluated condition is a deployable policy that includes its own failure behaviour. Passing the contract does not imply a correct translation.

IV. Experimental Protocol

Data. We use FLORES+ [2], [17] at a pinned revision: 997 `dev` and 1,012 `devtest` sentences per direction. The dataset's `fil_Latn` configuration maps to NLLB's `tgl_Latn`. Spanish is the benchmark's standard variety; we make no claim about heritage, dialectal, code-switched, or Los Angeles community Spanish. All prompt development, retrieval, and pilots used `dev` only.

Development pilot and protocol revision (development evidence). On the same 100 development IDs in all four directions (2,400 outcomes), the measured first contract asked reason-then-edit for one to six unique error categories with short notes. Llama fell back to the baseline on 227/400 such outputs (GPT-4o Mini: 8/400), almost entirely because of the diagnostic validators (e.g., repeated categories), not because of JSON parsing (399/400 parsed). Because a condition that mostly returns the baseline is not a clean strategy comparison, the principal investigator requested the minimal contract described in Section III, which we fixed together with a reliability gate (at most 20/400 fallbacks per model and 10/100 per direction) before rerunning the affected development conditions. Under the revised contract, neither editor fell back on any of its 400 development outputs (Llama 0/400, GPT-4o Mini 0/400), so the gate passed. Development quality estimates were descriptive and are reported only in the repository.

Freeze. Prompts, schemas, model revisions, decoding, retrieval policy, output contract, statistical families, thresholds, and code hashes were recorded in a freeze manifest before `devtest` was opened. Because the original split-generation code was not recovered, the splitter was reconstructed from the recorded specification and accepted after it reproduced all 2,991 saved development segmentations exactly; regenerated fragment translations matched 412/421 saved multi-sentence development items, the residual being FP16 beam-search numerics. On `devtest`, single-sentence items reuse the checksum-locked unsplit drafts, and every editor post-edits the same frozen S0 drafts. The matrix has 24 post-editing conditions (2 editors $\times$ 3 strategies $\times$ 4 directions) of 1,012 items each (24,288 outputs) plus the baseline.

Metrics and tests. COMET-22 (Unbabel/wmt22-comet-da, pinned checkpoint) is primary and reported as native score $\times 100$. BLEU (13a, mixed case, exponential smoothing) and chrF++ (character order 6, word order 2, $\beta{=}2$) are descriptive. For a comparison, d_i is the native segment COMET difference on the same sentence ID; we report $100\bar{d}$, a paired percentile 95% interval from 10,000 resamples, and a two-sided centred

paired-bootstrap p-value, $p = (1 + \#\{|\bar{d}_0^*| \geq |\bar{d}|\})/(B + 1)$. Holm correction is applied within two prespecified families of 24 tests: each condition against S0 (F1), and one-shot−zero-shot, reason−zero-shot, and reason−one-shot within each editor and direction (F2). Non-significant differences are inconclusive; an interval inside ± 0.5 points is flagged as practically small, not as equivalence. Pair and direction differences (RQ1, RQ4) are described from these estimates without separate tests.

Harm and over-editing. A segment improves if $d_i > 0.02$ (two reported points), is harmed if $d_i < -0.02$, and otherwise lies within the band. The protect pool contains items whose S0 COMET is at least 0.90 and the improve pool items below 0.80, both fixed on S0 before editing. A protect-pool over-edit is a textual change (NFC, outer whitespace removed) with $d_i < -0.02$, reported over all protect items, with sensitivity at 0.01 and 0.05. Digit, capitalized-token, language-identification (langdetect), length-ratio, and sentence-count screens are heuristics, not gold error labels.

V. Held-out Results

All 24 conditions contain the planned 1,012 items per direction, and every output passed the output contract: no condition used the baseline fallback (Table I). The measured effects are therefore editing behaviour, not contract failure. As a diagnostic outside the test families, sentence splitting itself raised S0 above the unsplit system by $+0.39$ (En→Tl), $+0.21$ (En→Es), and $+0.08$ points (Tl→En); Es→En was inconclusive.

Quality against the baseline (F1). GPT-4o Mini improves on S0 in every direction and strategy, by +0.26 to +1.00 points, all significant after Holm correction; five of these intervals, all in the Spanish pair, lie within ± 0.5 points, so several gains are clear but small. Llama 3 8B never improves significantly. Its zero-shot and one-shot outputs are significantly worse only for En→Tl (-0.48 and -0.39), with the other six intervals inside ± 0.5 points, and reason-then-edit is significantly worse in all four directions (-0.51 to -1.44). BLEU and chrF++ lie above S0 for all twelve GPT-4o Mini conditions and below it for all twelve Llama conditions, including those where COMET does not change significantly.

Pairs, baseline quality, and direction (RQ1, RQ4). GPT-4o Mini gains are larger for the Tagalog pair (+0.51 to +1.00) than for the Spanish pair (+0.26 to +0.53), and Llama losses are also largest for Tagalog. The Tagalog directions, however, contain more weak drafts (S0 COMET < 0.80: 183 and 82 items, against 104 and 60). Within this improve pool, GPT-4o Mini one-shot gains are similar across pairs (+3.11 to +5.29 points), while in the protect pool (S0 ≥ 0.90) every GPT-4o Mini condition stays near zero (-0.16 to $+0.15$). The pair difference thus tracks baseline quality as much as language. Direction effects are small and inconsistent: GPT-4o Mini gains more into English than out of English for every strategy in the Tagalog pair (e.g., $+0.83$ vs. $+0.64$ zero-shot), but not in the Spanish pair (one-shot $+0.36$ into English vs. $+0.53$ out of English).

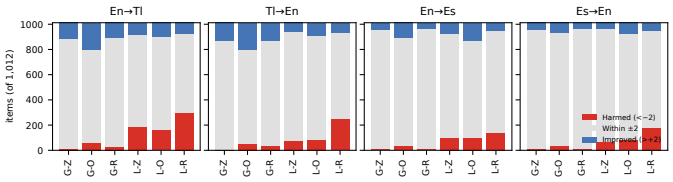


Fig. 1. Improved (>+2 points), within-band, and harmed (< −2) devtest items per condition (G/L: GPT-4o Mini/Llama; Z/O/R: zero-shot/one-shot/reason-then-edit). Unchanged outputs lie within the band.

Strategies (RQ2, F2). 15 of 24 strategy contrasts are significant after Holm correction. For GPT-4o Mini, the retrieved exemplar beats zero-shot in three directions (+0.17 to +0.31; Es→En inconclusive), and reason-then-edit falls below one-shot in the same three (-0.27 to -0.44) but below zero-shot only for Tl→En (-0.18). For Llama, one-shot and zero-shot do not differ significantly in any direction, whereas reason-then-edit is significantly worse than both in all four (-0.51 to -1.35).

Over-editing of strong drafts (RQ3). Edit rate alone does not predict harm (Fig. 1, Table II). GPT-4o Mini one-shot changes 3,249 of 4,048 drafts (80%) yet over-edits only 51 of 1,179 strong drafts (4.3%); its zero-shot and reason-then-edit rates are 0.8% and 2.1%. Llama edits at similar rates but over-edits 11.2% (zero-shot), 11.3% (one-shot), and 20.9% (reason-then-edit), up to 107 of 377 strong Tl→En drafts. Llama harms more segments than it improves in 9 of 12 conditions; GPT-4o Mini improves more than it harms in all 12. Every Llama strategy over-edits more strong drafts than every GPT-4o Mini strategy at the 0.01, 0.02, and 0.05 thresholds.

VI. Discussion

What separates the editors is not how often they edit. Both editors change most drafts under one-shot prompting, but GPT-4o Mini's changes to strong drafts are rarely harmful, whereas Llama's frequently are. Its losses concentrate where S0 is already good: in the protect pool, Llama's mean change ranges from -0.33 to -2.09 points across conditions, while its improve-pool changes are positive in 11 of 12 conditions. A post-editor is therefore only useful if it leaves correct drafts alone.

Explicit reasoning did not help. Reason-then-edit was never significantly better than zero-shot or one-shot, its point estimate was lower in all 16 such contrasts, and for Llama it was the most harmful strategy in every direction. The development pilot showed why this must be measured with a valid contract: under the original strict contract, 227 of 400 Llama reason-then-edit outputs fell back to the baseline, hiding the over-editing that the revised, fully valid condition exposes on held-out data. Diagnoses are model outputs, not verified explanations.

Preservation screens. Heuristic screens flag few items relative to S0: at most 23 outputs per condition change a digit (Llama zero-shot En→Tl), and language-ID mismatches rise above the baseline mainly for Llama reason-then-edit Tl→En (10 vs. 1). Screens are not gold errors; during

TABLE I

HELD-OUT FLORES+ DEVTEST RESULTS (1,012 ITEMS PER DIRECTION; ALL PLANNED OUTPUTS, INCLUDING BASELINE FALLBACK). C: COMET-22×100; Δ: POST-EDIT MINUS SENTENCE-SPLIT NLLB (S0) WITH PAIRED 95% BOOTSTRAP INTERVAL; MARKERS USE HOLM-ADJUSTED p OVER THE 24-TEST PRIMARY FAMILY ($^{\dagger}p < .05$, $^{\ddagger}p < .01$). FB: BASELINE FALLBACKS. BLEU/CHRF++ ARE DESCRIPTIVE.

		En→Tl			Tl→En			En→Es			Es→En		
Editor	Strategy	C	Δ [95% CI]	FB	C	Δ [95% CI]	FB	C	Δ [95% CI]	FB	C	Δ [95% CI]	FB
S0 split NLLB		85.20	–	–	87.57	–	–	86.67	–	–	87.37	–	–
GPT-4o Mini	Zero	85.84	+0.64‡ [+0.50, +0.77]	0	88.39	+0.83‡ [+0.69, +0.97]	0	86.95	+0.28‡ [+0.17, +0.39]	0	87.71	+0.34‡ [+0.24, +0.46]	0
	One	86.15	+0.95‡ [+0.78, +1.12]	0	88.57	+1.00‡ [+0.83, +1.18]	0	87.20	+0.53‡ [+0.37, +0.69]	0	87.73	+0.36‡ [+0.24, +0.50]	0
	Reason	85.71	+0.51‡ [+0.38, +0.64]	0	88.22	+0.65‡ [+0.50, +0.81]	0	86.93	+0.26‡ [+0.14, +0.38]	0	87.64	+0.27‡ [+0.17, +0.39]	0
Llama 3 8B	Zero	84.72	-0.48‡ [-0.65, -0.32]	0	87.47	-0.10 [-0.26, +0.05]	0	86.68	+0.00 [-0.18, +0.19]	0	87.36	-0.01 [-0.13, +0.12]	0
	One	84.81	-0.39‡ [-0.57, -0.21]	0	87.60	+0.03 [-0.12, +0.18]	0	86.85	+0.18 [-0.05, +0.40]	0	87.33	-0.04 [-0.24, +0.16]	0
	Reason	83.76	-1.44‡ [-1.70, -1.21]	0	86.25	-1.32‡ [-1.57, -1.07]	0	86.16	-0.51‡ [-0.71, -0.31]	0	86.63	-0.74‡ [-0.93, -0.55]	0

TABLE II

PROTECT-POOL OVER-EDITING ON DEVTEST: OVER-EDITED ITEMS / PROTECT-POOL ITEMS PER DIRECTION (PROTECT: S0 COMET≥0.90; OVER-EDIT: TEXT CHANGED AND NATIVE COMET LOSS >0.02).

Editor	Strategy	En→Tl	Tl→En	En→Es	Es→En
GPT-4o Mini	Zero	3/234	3/377	3/275	1/293
	One	6/234	26/377	7/275	12/293
	Reason	5/234	13/377	3/275	4/293
Llama 3 8B	Zero	42/234	35/377	29/275	26/293
	One	35/234	37/377	28/275	33/293
	Reason	57/234	107/377	32/275	51/293

development, source inspection found a numeric corruption that COMET rewarded and a source-consistent paraphrase that COMET penalized. **[PENDING: two or three source-reviewed held-out cases from the rule-based extraction (review_cases_PRIVATE.json)]**.

VII. LIMITATIONS AND ETHICAL CONSIDERATIONS

Two language pairs cannot identify a causal effect of resource level: language, direction, domain, and baseline quality are confounded. FLORES+ sources are largely English-original, so reverse directions involve translated source text. References are independent translations rather than minimal post-edits, so neither editing effort nor productivity is measured, and no bilingual human evaluation was conducted; COMET and the preservation screens are imperfect proxies. One open-weight checkpoint and one dated commercial snapshot do not represent all LLMs, pretraining exposure to the public benchmark cannot be ruled out, and deterministic settings do not guarantee identical remote behaviour. Fragment-level translation can lose cross-sentence context.

The study uses public benchmark text and automatic evaluation only. Results do not establish suitability for medical, legal, or other consequential translation, and standard-Spanish results should not be generalized to community or heritage varieties. LLM diagnoses can state unsupported errors confidently and are not expert explanations. Code, prompts, frozen configuration, and numeric results are released; benchmark text and generations are not redistributed. All GPT-4o Mini runs (development and held-out) cost about US$1.26 at list token rates; Llama ran on one A100. **[PENDING: venue-required AI-assistance and funding statements, to be confirmed by the PI]**

VIII. CONCLUSION

On held-out FLORES+ data, LLM post-editing of a strong, sentence-split NLLB-200 baseline helped or harmed depending above all on the editor. GPT-4o Mini improved every direction, most for the Tagalog pair, where weak drafts are more frequent, and most with one retrieved exemplar; Llama 3 8B never improved and over-edited strong drafts, especially with reason-then-edit. Gains came from weak drafts, direction effects were small, and explicit diagnosis did not improve either editor. Average gains, preservation of strong drafts, and output validity are separate properties that should all be measured before deployment. Future work should add bilingual human assessment, more language pairs and editors, and document-level context.

REFERENCES

[1] V. Raunak, A. Sharaf, Y. Wang, H. Awadalla, and A. Menezes, "Leveraging GPT-4 for automatic translation post-editing," in *Findings of the Association for Computational Linguistics: EMNLP 2023*. Singapore: Association for Computational Linguistics, Dec. 2023, pp. 12 009–12 024. [Online]. Available: https://aclanthology.org/2023.findings-emnlp.804/

[2] NLLB Team, "No language left behind: Scaling human-centered machine translation," arXiv:2207.04672, 2022. [Online]. Available: https://arxiv.org/abs/2207.04672

[3] B. Ahmadnia, S. Mohammed, S. Hooshmand, and L. Zhou, "Large language model-based automatic post-editing for test-time domain adaptation in neural machine translation," in *Proceedings of the International Conference on Artificial Intelligence (ICAI)*, 2026, to appear. Final proceedings metadata (publisher, pages, DOI) to be inserted when available; PI to confirm.

[4] D. Vilar, M. Freitag, C. Cherry, J. Luo, V. Ratnakar, and G. Foster, "Prompting PaLM for translation: Assessing strategies and performance," in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Toronto, Canada: Association for Computational Linguistics, Jul. 2023, pp. 15 406–15 427. [Online]. Available: https://aclanthology.org/2023.acl-long.859/

[5] D. Ki and M. Carpuat, "Guiding large language models to post-edit machine translation with error annotations," in *Findings of the Association for Computational Linguistics: NAACL 2024*. Mexico City, Mexico: Association for Computational Linguistics, Jun. 2024, pp. 4253–4273. [Online]. Available: https://aclanthology.org/2024.findings-naacl.265/

[6] M. V. Treviso, N. M. Guerreiro, S. Agrawal, R. Rei, J. Pombal, T. Vaz, H. Wu, B. Silva, D. van Stigt, and A. Martins, "xTower: A multilingual LLM for explaining and correcting translation errors," in *Findings of the Association for Computational Linguistics: EMNLP 2024*. Miami, Florida, USA: Association for Computational Linguistics, Nov. 2024, pp. 15 222–15 239. [Online]. Available: https://aclanthology.org/2024.findings-emnlp.892/

[7] D. Velazquez, M. Grace, K. Karageorgos, L. Carin, A. Schliem, D. Zaikis, and R. Wechsler, "LangMark: A multilingual dataset for automatic post-editing," in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, 2025, pp. 32 653–32 667. [Online]. Available: https://aclanthology.org/2025.acl-long.1569/

[8] F. Feng, Y. Yang, D. Cer, N. Arivazhagan, and W. Wang, "Language-agnostic BERT sentence embedding," in *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Dublin, Ireland: Association for Computational Linguistics, May 2022, pp. 878–891. [Online]. Available: https://aclanthology.org/2022.acl-long.62/

[9] R. Rei, C. Stewart, A. C. Farinha, and A. Lavie, "COMET: A neural framework for MT evaluation," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Online: Association for Computational Linguistics, Nov. 2020, pp. 2685–2702. [Online]. Available: https://aclanthology.org/2020.emnlp-main.213/

[10] R. Rei, J. G. C. de Souza, D. Alves, C. Zerva, A. C. Farinha, T. Glushkova, A. Lavie, L. Coheur, and A. F. T. Martins, "COMET-22: Unbabel-IST 2022 submission for the metrics shared task," in *Proceedings of the Seventh Conference on Machine Translation (WMT)*. Abu Dhabi, United Arab Emirates (Hybrid): Association for Computational Linguistics, Dec. 2022, pp. 578–585. [Online]. Available: https://aclanthology.org/2022.wmt-1.52/

[11] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, "Bleu: a method for automatic evaluation of machine translation," in *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics*. Philadelphia, Pennsylvania, USA: Association for Computational Linguistics, Jul. 2002, pp. 311–318. [Online]. Available: https://aclanthology.org/P02-1040/

[12] M. Popović, "chrF: character n-gram F-score for automatic MT evaluation," in *Proceedings of the Tenth Workshop on Statistical Machine Translation*. Lisbon, Portugal: Association for Computational Linguistics, Sep. 2015, pp. 392–395. [Online]. Available: https://aclanthology.org/W15-3049/

[13] ——, "chrF++: words helping character n-grams," in *Proceedings of the Second Conference on Machine Translation*. Copenhagen, Denmark: Association for Computational Linguistics, Sep. 2017, pp. 612–618. [Online]. Available: https://aclanthology.org/W17-4770/

[14] M. Post, "A call for clarity in reporting BLEU scores," in *Proceedings of the Third Conference on Machine Translation: Research Papers*. Brussels, Belgium: Association for Computational Linguistics, Oct. 2018, pp. 186–191. [Online]. Available: https://aclanthology.org/W18-6319/

[15] Meta, "Meta llama 3: Model card," 2024, original Llama 3 model release, 18 April 2024. [Online]. Available: https://github.com/meta-llama/llama3/blob/main/MODEL_CARD.md

[16] OpenAI, "GPT-4o mini: Advancing cost-efficient intelligence," 2024, 18 July 2024. [Online]. Available: https://openai.com/index/gpt-4o-mini-advancing-cost-efficient-intelligence/

[17] Open Language Data Initiative, "FLORES+: Multilingual machine translation benchmark," Dataset card, accessed 19 October 2026; experimental dataset revision is pinned separately. [Online]. Available: https://huggingface.co/datasets/openlanguagedata/flores_plus